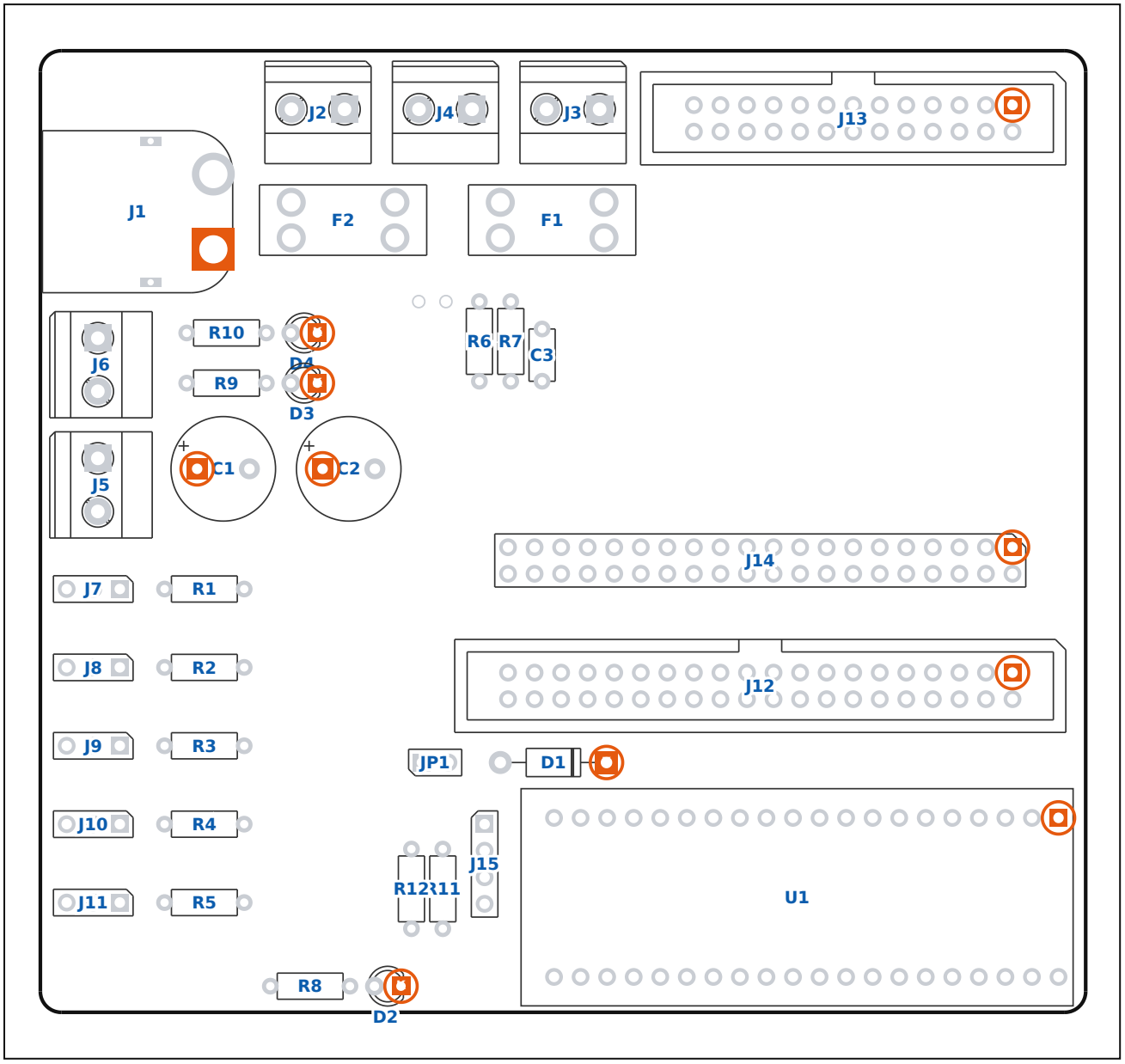
C.U.T.Y carrier v1.1 · Assembly drawing (top side)

PCB 100.1 × 92.1 mm, 2 layers, FR-4 1.6 mm · all parts through-hole, top side only · 39 placements, 225 THT joints · 0 SMD



Legend

- ◯ Orange pad = pad 1: **+** on C1/C2, **K** (cathode) on D1–D4, **pin 1** on J12/J13/J14/U1, **negative** on J1.
- Grey pads: all other pins. Blue text: reference designators.

Polarity and orientation

- **C1, C2** 1000 µF: + lead in the square pad (pad 1). Body max 20 mm tall.
- **D1** 1N5819: cathode band on the square pad side, as the silkscreen.
- **D2/D3/D4** 3 mm LEDs (green/blue/red): cathode (flat side, short lead) in the square pad. Seat flush.
- **J1** XT60PW-M (male): housing inside the silkscreen outline, mating face to the left board edge. Pad 1 = negative, pad 2 = battery +.
- **J2–J6** Phoenix terminal blocks: wire entries face the nearest board edge.
- **J12, J13** box headers: key slot toward the board edge marked on the silkscreen, pin 1 = square pad.
- **U1**: two 1×20 female sockets only (Black Pill is plugged in later). Leave the 4 SWD holes empty.
- **F1, F2**: fuse holders only, no fuses.
- **JP1**: fit the jumper shunt after soldering.

Assembly notes
Hand / selective solder is fine. All headers must stand perpendicular to the board (they mate with cables and a HAT). Clean flux residue. Substitutes with identical footprint and ratings are OK for resistors, C3, LEDs and pin headers; **do not substitute** J1, F1/F2 holders, J2–J6 terminal blocks or C1/C2 (height limit).

Bill of materials · turnkey

Files: cuty_carrier_PCBA_BOM.xlsx (BOM) · cuty_carrier_PCBA_CPL.xlsx (centroid, mm, origin = lower-left board corner, Y up)

Item #	Designator	Qty	Manufacturer	Mfg Part #	Description / Value	Your Instructions / Notes
1	C1,C2	2	Nichicon	UHE1C102MPD	Aluminum electrolytic 1000 uF 16 V, low ESR, D10 x 20 mm, 5.0 mm pitch	Polarized: + lead in the square pad (pad 1). Max body height 20 mm.
2	C3	1	Vishay	K104K15X7RF5TL2	Ceramic capacitor 100 nF 50 V X7R, radial 5.0 mm pitch	
3	D1	1	onsemi	1N5819RLG	Schottky diode 40 V 1 A, DO-41	Cathode band to the square pad (pad 1), as the silkscreen.
4	D2	1	Kingbright	WP710A10SGD	LED 3 mm green	Cathode (flat side, short lead) in the square pad (pad 1). Flush with the board.
5	D3	1	Kingbright	WP710A10QBC/D	LED 3 mm blue	Cathode (flat side, short lead) in the square pad (pad 1). Flush with the board.
6	D4	1	Kingbright	WP710A10ID	LED 3 mm red	Cathode (flat side, short lead) in the square pad (pad 1). Flush with the board.
7	F1,F2	2	Keystone Electronics	3568	Fuse holder, mini blade (ATM), PCB mount, 4 pins	Holder only. Do NOT insert fuses (owner inserts 10 A in F1 and 5 A in F2).
8	J1	1	Changzhou Amass	XT60PW-M	XT60 male, right angle, PCB mount (LCSC C98732)	Male. Housing exactly inside the silkscreen outline: pad 1 (square) = negative, pad 2 = positive (battery +). Do not substitute.
9	J2,J3,J4,J5,J6	5	Phoenix Contact	1715721	Terminal block MKDS 1,5/2-5,08, 2 pos, 5.08 mm	Wire entries face the nearest board edge, as the silkscreen outline. Do not substitute pitch.
10	J7,J8,J9,J10,J11	5	Sullins	PREC003SAAN-RC	Pin header male 1x3, 2.54 mm, vertical	Can be cut from a breakaway strip.
11	J15	1	Sullins	PREC004SAAN-RC	Pin header male 1x4, 2.54 mm, vertical	Can be cut from a breakaway strip.
12	JP1	1	Sullins	PREC002SAAN-RC	Pin header male 1x2, 2.54 mm, vertical	Can be cut from a breakaway strip.
13	JP1 (shunt)	1	Sullins	SPC02SYAN	Jumper shunt 2.54 mm	Fit the shunt on JP1 after soldering.
14	J12	1	Wurth Elektronik	61204021621	Box header (shrouded) 2x20, 2.54 mm, vertical	Keyed: key slot and pin 1 (square pad) exactly as the silkscreen.
15	J13	1	Wurth Elektronik	61202621621	Box header (shrouded) 2x13, 2.54 mm, vertical	Keyed: key slot and pin 1 (square pad) exactly as the silkscreen.
16	J14	1	Sullins	PRPC020DAAN-RC	Pin header male 2x20, 2.54 mm, vertical	
17	U1	2	Sullins	PPTC201LFBN-RC	Female socket 1x20, 2.54 mm, 8.5 mm tall	Two 1x20 female sockets in the two 20-pin rows. Leave the 4 SWD holes (marked SWD) empty. Do not install any module.
18	R1,R2,R3,R4,R5	5	Yageo	MFR-25FBF52-220R	Resistor 220 ohm 1% 1/4 W metal film, axial	
19	R6	1	Yageo	MFR-25FBF52-100K	Resistor 100 kohm 1% 1/4 W metal film, axial	
20	R7	1	Yageo	MFR-25FBF52-22K	Resistor 22 kohm 1% 1/4 W metal film, axial	
21	R8,R9	2	Yageo	MFR-25FBF52-1K	Resistor 1 kohm 1% 1/4 W metal film, axial	
22	R10,R11,R12	3	Yageo	MFR-25FBF52-4K7	Resistor 4.7 kohm 1% 1/4 W metal film, axial	

Not assembled (supplied and fitted by the owner)

WeAct Black Pill STM32F401CCU6 module · mini blade fuses 10 A (F1) and 5 A (F2) · 4 × M3 standoffs (H1–H4)